CSE 4224: Digital System Design

Project Title: ATM Machine Using Basys 3 Board

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1. Introduction

Automated Teller Machines (ATMs) are among the most widely used electronic systems in the financial sector, providing users with secure and convenient access to banking services. The fundamental operations of an ATM, including user authentication, balance inquiry, cash withdrawal, and deposit, rely on core digital system concepts such as finite state machines, arithmetic logic units, memory management, and input/output interfacing.

This project presents the design and implementation of a simplified ATM machine using the Basys 3 FPGA development board. The system is developed using Verilog hardware description language and synthesized using Xilinx Vivado 2023. The Basys 3 board, which features an Artix-7 FPGA (xc7a35tcpg236-1), provides the necessary hardware resources including switches, push buttons, LEDs, and seven-segment displays to create a fully functional ATM simulation.

The system supports multiple users with individual PINs and account balances, secure PIN authentication with a three-attempt lockout mechanism, deposit and withdrawal operations with real-time balance verification, and a 30-second timeout for automatic session termination. This project demonstrates the practical application of digital system design principles including FSM-based control flow, combinational and sequential logic design, memory interfacing, and hardware-software co-design on a real FPGA platform.

2. Objectives

The primary objectives of this project are as follows:

First, to design and implement an FSM-based control system that manages all ATM operations through well-defined states and transitions, ensuring reliable and predictable system behavior.

Second, to develop a multi-functional Arithmetic Logic Unit (ALU) capable of performing addition for deposits, subtraction for withdrawals, comparison for sufficient fund verification, and equality checking for PIN authentication.

Third, to integrate secure authentication and memory management by implementing a multi-user memory system that stores PINs and balances, along with a PIN verification mechanism that locks out users after three consecutive failed attempts.

Fourth, to create a user-friendly interface using Basys 3 resources, including switches for data input, buttons for navigation and confirmation, LEDs for status indication, and seven-segment displays for numeric and text output.

Fifth, to implement security features including a 30-second inactivity timeout that automatically terminates the session and ejects the card.

3. Methodology

The ATM system is designed using a modular approach where each functional block is implemented as a separate Verilog module. The modules communicate through well-defined interfaces controlled by the central finite state machine. The entire system is synthesized and implemented on the Basys 3 FPGA board.

3.1 System Architecture

The system follows a modular architecture consisting of seven interconnected components: the Control Unit (FSM), ALU, Memory Module, Button Debouncer, Seven Segment Display Controller, Binary to BCD Converter, and Timeout Timer. The Control Unit serves as the central coordinator, generating control signals for all other modules based on the current state and user inputs.

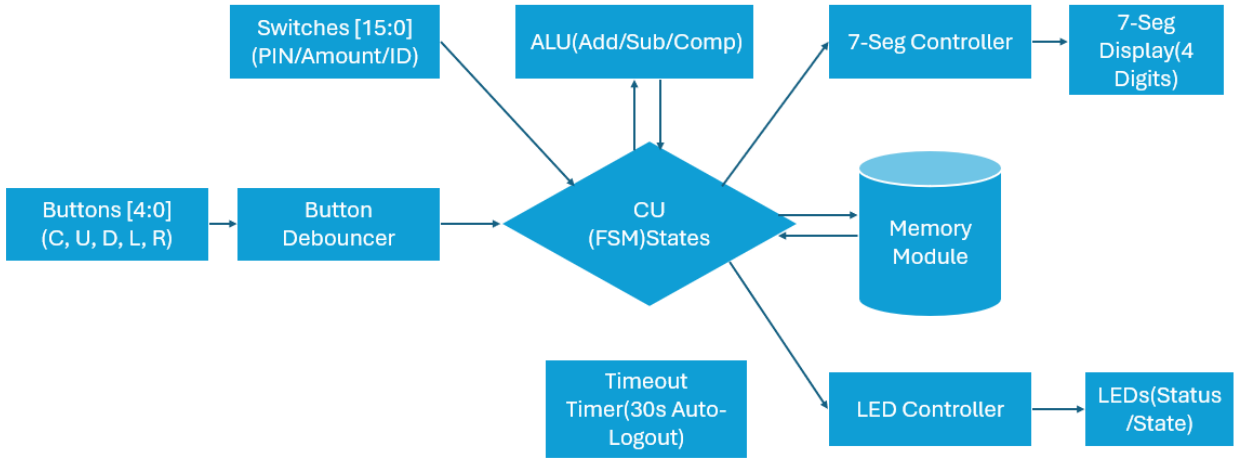


Figure 1: System Architecture

System Architecture - Block diagram showing the interconnection between Control Unit, ALU, Memory Module, Button Debouncer, 7-Segment Display Controller, Timeout Timer, and Basys 3 I/O interfaces.

The top-level module (atm_top) instantiates all sub-modules and establishes the signal connections between them. A power-on reset generator within the top module ensures proper initialization of all components upon FPGA configuration.

3.2 Data Flow

The data flow through the system follows a well-defined path from user input to output display. When a user interacts with the system through switches and buttons, the debounced signals are received by the Control Unit. The Control Unit then coordinates with the Memory Module to read or write data and with the ALU to perform arithmetic or comparison operations. The results are displayed through the seven-segment display and LEDs.

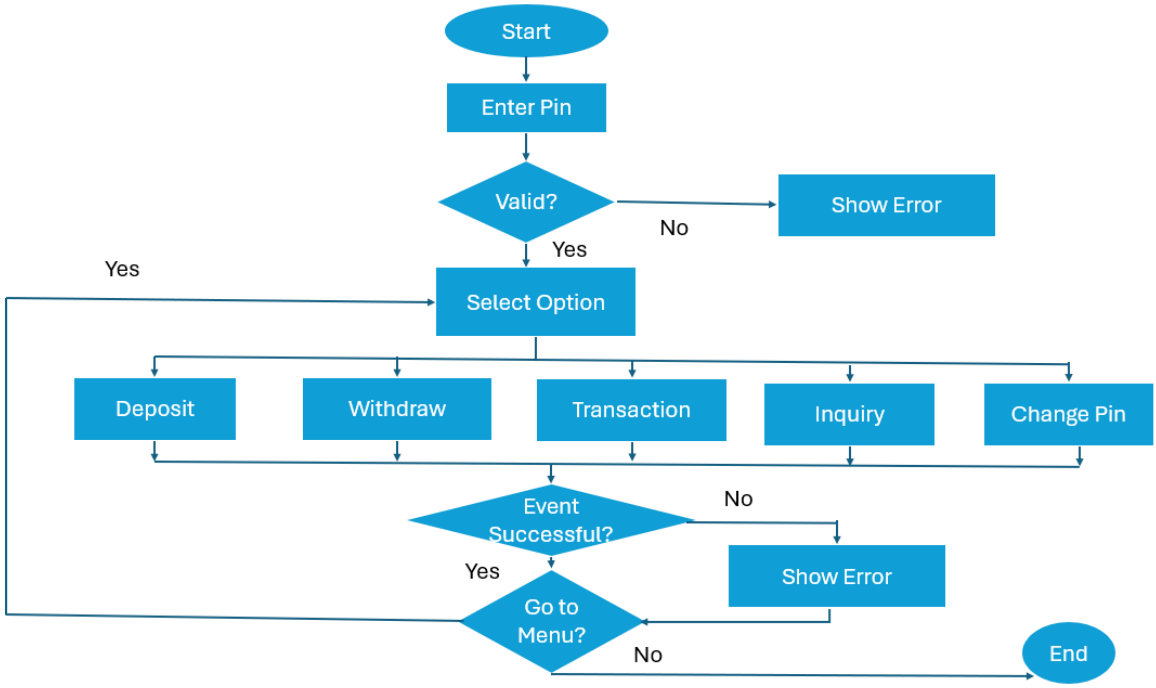


Figure 2: Data Flow Diagram

Data Flow Diagram - Flowchart showing the complete transaction flow from card insertion through PIN verification, menu selection, transaction processing, and session termination.

3.3 Module Description

3.3.1 Control Unit (FSM)

The Control Unit is the brain of the ATM system, implemented as a finite state machine with 14 states. It manages all system states from IDLE through PIN entry, verification, menu selection, transaction processing, and card ejection. The FSM generates control signals including mem_read and mem_write for memory access, alu_op for ALU operation selection, display_mode for seven-segment display control, and led_out for status LED management.

The 14 states of the FSM are: IDLE (state 0), USER_SELECT (state 1), PIN_ENTRY (state 2), PIN_VERIFY (state 3), PIN_RESULT (state 4), MAIN_MENU (state 5), CHECK_BALANCE (state 6), WITHDRAW (state 7), DEPOSIT (state 8), TRANSACTION (state 9), DISPENSE (state 10), ERROR (state 11), SUCCESS (state 12), and EJECT_CARD (state 13).

The Control Unit evaluates ALU flags (zero_flag for equality, negative_flag for insufficient funds) and user button inputs to determine state transitions. It also monitors the timeout signal from the Timeout Timer and forces an automatic card ejection when the timeout expires during any active state.

A three-attempt PIN lockout mechanism is implemented within the Control Unit. A pin_attempts counter tracks consecutive failed PIN entries, and upon reaching three failures, the system transitions directly to the EJECT_CARD state without allowing further attempts.

3.3.2 Arithmetic Logic Unit (ALU)

The ALU is a purely combinational module that performs four operations based on a 3-bit operation code. The NOP operation (code 000) produces no output. The ADD operation (code 001) computes the sum of operands A and B, used during deposit transactions to calculate the new balance. The SUB operation (code 010) computes the difference of A minus B, used during withdrawal transactions, with clamping to zero if the result would be negative. The COMPARE operation (code 011) checks whether operand A is greater than or equal to operand B, used to verify sufficient funds before a withdrawal. The EQUAL operation (code 100) checks equality between operands A and B, used for PIN verification.

The ALU produces two flag outputs: the zero_flag, which is asserted when the result equals zero or when operands are equal during an equality check, and the negative_flag, which is asserted when operand A is less than operand B, indicating insufficient funds.

3.3.3 Memory Module

The Memory Module is implemented as a 16-location by 16-bit register array with synchronous read and write operations. The memory is organized to store data for four users, with each user occupying two consecutive addresses: the even address stores the PIN and the odd address stores the account balance.

The default values loaded upon reset are: User 0 with PIN 1234 and balance 5000 at addresses 0 and 1, User 1 with PIN 5678 and balance 3000 at addresses 2 and 3, User 2 with PIN 1111 and balance 8000 at addresses 4 and 5, and User 3 with PIN 9999 and balance 1500 at addresses 6 and 7. Addresses 8 through 15 are reserved for transaction history and are initialized to zero.

Memory operations are controlled by the mem_read and mem_write signals generated by the Control Unit. Write operations update the balance after successful deposit or withdrawal transactions.

3.3.4 Button Debouncer

The Button Debouncer module eliminates mechanical switch bounce by requiring a stable input signal for approximately 10 milliseconds (1,000,000 clock cycles at 100 MHz) before registering a change. A two-stage synchronizer is used at the input to prevent metastability issues when sampling asynchronous button signals.

The module also includes rising edge detection logic, producing a single clock cycle pulse when a valid button press is detected. Five instances of this module are instantiated in the top-level design, one for each of the five push buttons on the Basys 3 board.

3.3.5 Seven Segment Display Controller

The Seven Segment Display Controller drives the four-digit seven-segment display on the Basys 3 board using time-division multiplexing at approximately 1 kHz. The controller supports nine display modes: blank (mode 0), PIN entry masking showing four dashes (mode 1), numeric balance display (mode 2), numeric amount entry (mode 3), menu text display (mode 4), error message "Err" (mode 5), success message "donE" (mode 6), idle message "IdLE" (mode 7), and user ID selection display (mode 8).

The display mode is selected by the Control Unit based on the current FSM state. The Binary to BCD Converter is used to convert 16-bit binary values to individual decimal digits for numeric display modes.

3.3.6 Binary to BCD Converter

The Binary to BCD Converter uses the Double Dabble algorithm to convert a 16-bit binary number (range 0 to 9999) into four BCD digits. This is a purely combinational module that performs 16 shift-and-add-3 iterations to produce the thousands, hundreds, tens, and ones digits. These digits are then used by the Seven Segment Display Controller for numeric display.

3.3.7 Timeout Timer

The Timeout Timer implements a 30-second inactivity counter that automatically terminates the user session for security purposes. The timer uses a 32-bit counter that counts up to 2,999,999,999 at the 100 MHz clock frequency. The timer is reset whenever the user performs any action (button press), and the expired signal is asserted when the

counter reaches its maximum value. The Control Unit monitors this signal and transitions to the EJECT_CARD state upon expiration.

3.4 Basys 3 Interface Mapping

The physical interface mapping on the Basys 3 board is as follows:

For inputs, switches SW[15:0] are used for entering PIN values, transaction amounts, and user IDs. The center button (btnC) serves as the confirm or enter button. The up button (btnU) is used to navigate upward in the menu. The down button (btnD) navigates downward. The left button (btnL) functions as cancel or back. The right button (btnR) simulates card insertion.

For outputs, the four-digit seven-segment display shows PIN entry masks, account balances, transaction amounts, menu options, and status messages. The 16 LEDs provide visual feedback for the current system state, transaction status, and error conditions. Each FSM state activates a specific LED pattern, making it easy to identify the current operational state during demonstration.

3.5 FSM State Descriptions

The IDLE state is the initial state where the system waits for card insertion, displaying "IdLE" on the seven-segment display with LED[0] illuminated. Upon pressing btnR, the system transitions to USER_SELECT.

In the USER_SELECT state, the user sets their ID (0 to 3) on switches SW[1:0] and confirms with btnC. The display shows the selected user ID.

During PIN_ENTRY, the display shows four dashes to mask the input while the user sets the PIN value on the switches and confirms with btnC. The entered value is captured and the system proceeds to PIN_VERIFY.

The PIN_VERIFY state reads the stored PIN from memory and uses the ALU's equality operation to compare it with the entered PIN. If the PINs match, the system reads the account balance and transitions to MAIN_MENU. If they do not match, the pin_attempts counter increments, and the system either returns to PIN_ENTRY for another attempt or transitions to EJECT_CARD after three failures.

In MAIN_MENU, the user navigates between four options using btnU and btnD: Check Balance (bAL), Deposit (dEP), Withdraw (Udr), and Exit. The selected option is confirmed with btnC.

CHECK_BALANCE displays the current account balance on the seven-segment display and returns to the menu upon button press.

During WITHDRAW, the user enters the withdrawal amount on the switches. The ALU first compares the balance with the requested amount. If sufficient funds exist, the ALU subtracts the amount from the balance, the new balance is written to memory, and the system proceeds to DISPENSE. If funds are insufficient, the system transitions to ERROR.

The DEPOSIT state accepts an amount from the switches, uses the ALU to add it to the current balance, writes the new balance to memory, and transitions to TRANSACTION.

TRANSACTION briefly displays the updated balance before proceeding to either DISPENSE for withdrawals or SUCCESS for deposits.

DISPENSE simulates cash dispensing with an LED animation pattern for approximately one second before transitioning to SUCCESS.

SUCCESS displays "donE" for approximately one second and then returns to MAIN_MENU.

ERROR displays "Err" for approximately one second and then returns to the appropriate state based on the error type.

EJECT_CARD displays "IdLE" briefly with a distinctive LED pattern before returning to the IDLE state.

3.6 Memory Organization

The memory is organized as a simple register file with the following address mapping:

Address 0: User 0 PIN (default: 1234)

Address 1: User 0 Balance (default: 5000)

Address 2: User 1 PIN (default: 5678)

Address 3: User 1 Balance (default: 3000)

Address 4: User 2 PIN (default: 1111)

Address 5: User 2 Balance (default: 8000)

Address 6: User 3 PIN (default: 9999)

Address 7: User 3 Balance (default: 1500)

Addresses 8-15: Reserved for transaction history

The address calculation for any user follows a simple formula: PIN address equals user_id multiplied by 2, and balance address equals user_id multiplied by 2 plus 1.

4. Simulation and Results

The system was verified through behavioral simulation in Vivado 2023 using a comprehensive testbench. The testbench instantiates the complete atm_top module and simulates realistic button press sequences by holding button signals high for 15 milliseconds to satisfy the debounce timing requirements.

Ten test scenarios were executed to validate all system functionalities:

Test 1 verified card insertion by pressing btnR, confirming the system transitions from IDLE to USER_SELECT state with appropriate LED indication.

Test 2 verified user selection by setting switches to User 0 and confirming, with the system transitioning to PIN_ENTRY.

Test 3 validated correct PIN authentication by entering PIN 1234 for User 0, confirming successful verification and transition to MAIN_MENU.

Test 4 confirmed the check balance functionality, verifying that the display shows the correct initial balance of 5000 for User 0.

Test 5 tested the deposit operation by depositing 500, with the ALU correctly computing the new balance and the memory being updated accordingly.

Test 6 verified the updated balance after deposit, confirming the display shows 5500.

Test 7 tested the withdrawal operation by withdrawing 200, verifying proper fund checking, balance computation, and the dispense animation sequence.

Test 8 confirmed the exit functionality, with the system properly ejecting the card and returning to IDLE.

Test 9 validated the PIN lockout mechanism by entering three consecutive wrong PINs, confirming the system displays an error and ejects the card after the third failure.

Test 10 tested the insufficient funds scenario by attempting to withdraw 9000 from User 3's account with only 1500 balance, confirming the system properly detects the condition through the ALU's negative flag and displays an error.

All ten tests completed successfully, demonstrating correct operation of the FSM state transitions, ALU computations, memory read/write operations, display outputs, and security features.

5. Discussion

The ATM machine implementation successfully demonstrates the integration of multiple digital system components on a single FPGA platform. The modular design approach proved effective in managing the complexity of the system, as each module could be designed, tested, and verified independently before integration.

The FSM-based control architecture provides a clear and maintainable structure for managing the complex sequence of operations required in an ATM transaction. The 14-state design covers all necessary operational modes while maintaining readability and ease of modification. The use of sub-steps within certain states, such as PIN_VERIFY and WITHDRAW, allows multi-cycle operations to be handled within the synchronous FSM framework without introducing additional states.

The ALU design, while simple compared to a general-purpose processor ALU, effectively handles all required operations for the ATM application. The use of flag outputs (zero_flag and negative_flag) provides an efficient mechanism for the Control Unit to evaluate operation results without needing to examine the full result value.

The memory module uses a register-based implementation rather than block RAM, which is appropriate for the small memory requirements of this application. The synchronous read and write operations ensure reliable data access within the clocked system. The memory organization with interleaved PIN and balance storage for each user provides straightforward address calculation using simple bit manipulation.

The button debouncer design addresses a critical practical concern in FPGA-based systems with mechanical switch inputs. The 10-millisecond debounce period effectively eliminates spurious transitions, and the two-stage synchronizer prevents metastability issues that could cause unpredictable behavior.

The timeout security feature adds an important layer of protection against unauthorized access. In a real-world scenario, a user who walks away from an ATM without completing their session would be automatically logged out after 30 seconds of inactivity.

One limitation of the current implementation is the use of 16-bit values for PIN and balance storage, which limits balances to a maximum of 65,535 and PINs to numeric values within this range. The seven-segment display further limits the displayable range to 0-9999. Additionally, the system does not implement encryption for PIN storage, which would be essential in a production ATM system. The PIN values are stored in plain form in the register file, which is acceptable for this educational demonstration but would not meet security standards for real financial applications.

Another consideration is that the system uses a power-on reset mechanism, meaning the memory contents including all balances are reset to their default values each time the FPGA is reprogrammed or power-cycled. In a real system, non-volatile storage would be required to persist account data across power cycles.

6. Conclusion

This project successfully demonstrates the design and implementation of a simplified ATM machine on the Basys 3 FPGA board using Verilog HDL. The system integrates fundamental digital design concepts including finite state machine design, arithmetic logic unit implementation, memory management, input debouncing, display multiplexing, and security timer mechanisms.

The modular architecture consisting of seven distinct Verilog modules provides a clean separation of concerns and facilitates understanding of how individual digital components work together to form a complete system. The FSM-based control approach effectively manages the complex operational flow of an ATM, from user authentication through transaction processing to session termination.

All design objectives were met, including multi-user support with four pre-configured accounts, secure PIN authentication with three-attempt lockout, deposit and withdrawal operations with real-time balance verification, user-friendly interface using all available Basys 3 I/O resources, and automatic timeout security. The system was verified through comprehensive behavioral simulation covering normal operations, edge cases, and error conditions.

This project serves as a practical demonstration of how theoretical digital system design concepts are applied in the development of real-world embedded systems, bridging the gap between classroom learning and hardware implementation.

7. References

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